

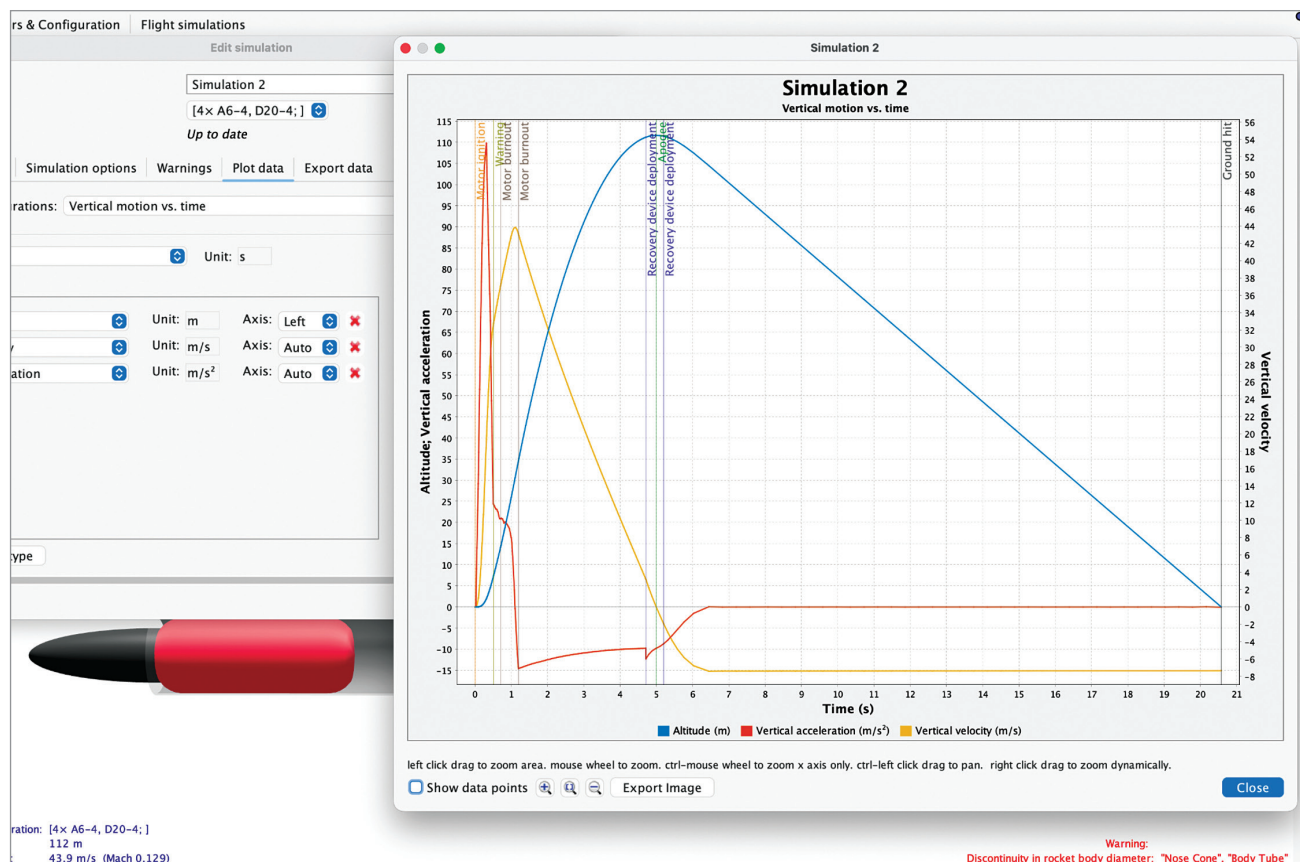


Fig. 4: OpenRocket flight simulation results

the centre, and the parachute system is usually located toward the front section of the body tube. Proper component arrangement is necessary to maintain the correct centre of gravity and ensure stable flight behaviour.

From an aerodynamic perspective, the body tube surface should be smooth and properly aligned with the nose cone and fins. One important parameter in rocket staging is the mass ratio, given by:

$$\text{Mass ratio} = m_0/m_f$$

where m_0 is the initial mass and m_f is the final mass after fuel consumption. A higher mass ratio generally improves performance and altitude capability, but it also increases structural and design complexity.

Multi-stage rocket. In multi-stage systems, proper alignment between stages becomes extremely important. Even small angular misalignments can destabilise the

rocket during separation or ignition. The timing of stage ignition is also critical. Delayed ignition can cause a loss of momentum, while premature ignition can result in structural damage to the rocket.

Another important challenge in multi-stage rockets is that the centre of gravity changes dynamically during stage separation. Therefore, stability must be ensured not only before launch but also after each stage separates during flight. Fig. 3 shows an initial model rocket system design created in OpenRocket, showing the nose cone, body tube, CG, and CP.

Fin design. After the body tube, one of the most important parts of rocket design is the fin system. Fins are responsible for stabilising the rocket during flight and helping it maintain direction. The main purpose of fins is to move the CP behind the CG, allowing the rocket to naturally align itself with the

direction of airflow during flight. Because of this, fin size, shape, placement, thickness, and alignment become extremely critical for rocket stability.

One of the first rules in fin design is choosing the number of fins. Most model rockets use either three or four fins, but three fins are generally preferred because they provide good stability with lower aerodynamic drag and reduced weight. Using too many fins increases drag unnecessarily, while using very few fins may reduce stability.

The fin size is generally determined relative to the rocket's body diameter. A common design rule is:

Root chord = 1 to 1.5 × body diameter

Fin height = 1 to 2 × body diameter

Tip chord = 0.3 to 0.6 × root chord

These ratios help maintain a balance between stability and drag. Oversized fins increase drag and